

Generalized lushness passes to all Bochner L_∞ spaces

A full answer to one open direction in arXiv:2007.10589

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Status. Candidate full solution, likely valid, pending human review. If a real Banach space X is generalized-lush, then $L_\infty(\mu, X)$ is generalized-lush for every measure space. This resolves the previously unknown forward L_∞ implication in the source. The three converse questions stated beside it are not claimed here.

1 Source question

For $x^* \in S_{X^*}$ and $0 < \alpha < 1$, write

$$S_X(x^*, \alpha) = \{x \in B_X : x^*(x) > 1 - \alpha\}.$$

A Banach space X is *generalized-lush* (GL) if for every $x \in S_X$ and $\alpha > 0$ there is $x^* \in S_{X^*}$ such that $x \in S_X(x^*, \alpha)$ and

$$\text{dist}(y, S_X(x^*, \alpha)) + \text{dist}(y, -S_X(x^*, \alpha)) < 2 + \alpha \quad (y \in S_X).$$

Zhao and Tan record that GL is known to pass from X to $C(K, X)$ and $L_1(\mu, X)$, but explicitly state that the corresponding assertion for $L_\infty(\mu, X)$ was unknown [1].

A PROPERTY IN VECTOR-VALUED FUNCTION SPACES

3

Let us make a comment here on vector-valued function spaces for GL-spaces. We only know that if X is a GL-space, then so are $C(K, X)$ ([16, Theorem 2.10]) and $L_1(\mu, X)$ ([10, Theorem 5.1]). It is not known whether this is true for $L_\infty(\mu, X)$ nor if X is a GL-space whenever $C(K, X)$, $L_1(\mu, X)$ or $L_\infty(\mu, X)$ is a GL-space, where X is non-separable.

Throughout the paper, given a compact Hausdorff topological space K and a Banach space X , $C(K, X)$ is the Banach space of all continuous functions from K into X endowed with the supremum norm. Given a σ -finite measure space (Ω, Σ, μ) , for $A \in \Sigma$, χ_A is the characteristic function of A , and for a Banach space X , $L_\infty(\mu, X)$ is the Banach space of all (classes of) measurable functions f from Ω into X which are essentially bounded, endowed with the essential supremum norm

$$\|f\|_\infty = \text{ess sup}\{\|f(t)\| : t \in \Omega\}.$$

$L_1(\mu, X)$ is the Banach space of all (classes of) Bochner-integrable functions from Ω into X endowed with the integral norm

2 Proof intuition

Fix $f \in S_{L_\infty(\mu, X)}$. Choose a Boolean ultrafilter concentrating on a set where $\|f(t)\|$ is almost one. The normalized values of f have separable range, so approximate them uniformly by a

countably valued map. At each of those countably many values choose a GL slice functional on X . Taking the ultrafilter limit of the resulting scalar pairings defines one functional on $L_\infty(\mu, X)$ whose slice contains f .

For an arbitrary g , countable approximation again makes pointwise GL witnesses measurable. Their two pointwise distances have sum below $2 + o(1)$, but taking two separate essential suprema could destroy that bound. The key step is to divide the first distance into finitely many short intervals. The ultrafilter selects one interval. Patch the witnesses only on that selected region and leave g unchanged elsewhere. On the selected region the first distance has almost no oscillation, so the two global suprema still add to less than $2 + \varepsilon$.

3 Two elementary lemmas

Lemma 1 (The GL estimate holds on the ball). *Let S be a slice of B_X . If*

$$D(y) := \text{dist}(y, S) + \text{dist}(y, -S) < c \quad (y \in S_X),$$

then $D(y) < c$ for every $y \in B_X$.

Proof. The function D is convex and even. If $y = rz$ with $z \in S_X$ and $0 \leq r \leq 1$, then

$$y = \frac{1+r}{2}z + \frac{1-r}{2}(-z).$$

Convexity and evenness give $D(y) \leq D(z) < c$. □

Let $\mathfrak{B} = \Sigma/\mathcal{N}_\mu$ be the measure algebra. If $\mathcal{U}$ is an ultrafilter on $\mathfrak{B}$ and q is a bounded real measurable function, $\lim_{\mathcal{U}} q$ denotes its ultrafilter limit. It is unchanged by null modifications and preserves sums and scalar multiples.

Lemma 2 (A countably valued field gives a functional). *Suppose $\phi : \Omega \rightarrow S_{X^*}$ is constant on each member of a countable measurable partition. Then*

$$F_0(h) := \lim_{\mathcal{U}} \phi(t)(h(t)) \quad (h \in L_\infty(\mu, X))$$

is a well-defined linear functional of norm at most one. If $[E] \in \mathcal{U}$ and $\phi(t)(h(t)) \geq c$ almost everywhere on E , then $F_0(h) \geq c$. The analogous upper estimate also holds.

Proof. For strongly measurable h , the scalar function $t \mapsto \phi(t)(h(t))$ is measurable because ϕ is countably valued. It is bounded in absolute value by $\|h\|_\infty$. Linearity and the norm bound follow from the corresponding properties of ultrafilter limits. The last assertion follows because an ultrafilter limit ignores the complement of every member of $\mathcal{U}$. □

4 The full L_∞ stability theorem

Theorem 3. *Let (Ω, Σ, μ) be a nonzero measure space and let X be a real Banach space. If X is generalized-lush, then the Bochner space $L_\infty(\mu, X)$ is generalized-lush.*

Proof. Fix $f \in S_{L_\infty(\mu, X)}$ and $0 < \varepsilon < 1$, and put $\delta = \varepsilon/20$. The measurable set

$$A = \{t : \|f(t)\| > 1 - \delta\}$$

is non-null. Extend the filter generated by $[A]$ to an ultrafilter $\mathcal{U}$ on $\mathfrak{B}$.

On A , the map $u(t) = f(t)/\|f(t)\|$ is strongly measurable and has separable range. Hence there are a measurable partition $A = \bigcup_{n \geq 1} A_n$ and points $x_n \in S_X$ such that

$$\|u(t) - x_n\| < \delta \quad (t \in A_n). \tag{1}$$

For every n , generalized lushness supplies $\phi_n \in S_{X^*}$ such that

$$\phi_n(x_n) > 1 - \delta \quad (2)$$

and, with $S_n = S_X(\phi_n, \delta)$,

$$\text{dist}(y, S_n) + \text{dist}(y, -S_n) < 2 + \delta \quad (y \in B_X), \quad (3)$$

where Lemma 1 was used for the ball.

Define $\phi(t) = \phi_n$ on A_n and give it any fixed value in S_{X^*} off A . Lemma 2 defines $F_0 \in L_\infty(\mu, X)^*$ with $\|F_0\| \leq 1$. On A_n ,

$$\phi(t)(f(t)) = \|f(t)\|\phi_n(u(t)) > (1 - \delta)(1 - 2\delta) > 1 - 3\delta$$

by (1) and (2). Since $[A] \in \mathcal{U}$,

$$F_0(f) \geq 1 - 3\delta > 0. \quad (4)$$

Thus $c := \|F_0\| > 0$; put $F = F_0/c \in S_{L_\infty(\mu, X)^*}$. Because $c \leq 1$, (4) gives

$$F(f) > 1 - \varepsilon.$$

It remains to verify the distance condition for an arbitrary $g \in S_{L_\infty(\mu, X)}$. Strong measurability gives a countably valued measurable map $s(t) = y_m$ on a partition $(C_m)_{m \geq 1}$, with $y_m \in B_X$ and

$$\|g(t) - s(t)\| < \delta \quad \text{almost everywhere.} \quad (5)$$

For each pair (n, m) , (3) permits choices $a_{nm} \in S_n$ and $b_{nm} \in -S_n$ satisfying

$$\|y_m - a_{nm}\| + \|y_m - b_{nm}\| < 2 + 2\delta. \quad (6)$$

On $A_n \cap C_m$ set $a(t) = a_{nm}$ and $b(t) = b_{nm}$. These maps are strongly measurable, take values in B_X , and satisfy

$$\phi(t)(a(t)) > 1 - \delta, \quad \phi(t)(b(t)) < -1 + \delta, \quad (7)$$

$$d_+(t) + d_-(t) < 2 + 4\delta, \quad d_\pm(t) := \|g(t) - a(t)\|, \quad \|g(t) - b(t)\|, \quad (8)$$

on A , by (5) and (6).

Partition A into finitely many measurable sets according to which interval of length δ contains $d_+(t)$; the last interval is truncated at two. Exactly one member E of this finite partition has $[E] \in \mathcal{U}$. For some integer $j \geq 0$ we therefore have on E

$$j\delta \leq d_+(t) < (j+1)\delta, \quad d_-(t) < 2 + 4\delta - j\delta.$$

Define

$$h_+(t) = \begin{cases} a(t), & t \in E, \\ g(t), & t \notin E, \end{cases} \quad h_-(t) = \begin{cases} b(t), & t \in E, \\ g(t), & t \notin E. \end{cases}$$

Both functions lie in the unit ball. Since $[E] \in \mathcal{U}$, Lemma 2 and (7) give

$$F_0(h_+) \geq 1 - \delta, \quad F_0(h_-) \leq -1 + \delta.$$

Dividing by $c \leq 1$ only improves these signed inequalities, so

$$h_+ \in S_{L_\infty}(F, \varepsilon), \quad h_- \in -S_{L_\infty}(F, \varepsilon).$$

Finally, the patching and the selected distance bin give

$$\begin{aligned} \|g - h_+\|_\infty + \|g - h_-\|_\infty &\leq (j+1)\delta + (2 + 4\delta - j\delta) \\ &= 2 + 5\delta < 2 + \varepsilon. \end{aligned}$$

The same functional F works for every g ; hence $L_\infty(\mu, X)$ is GL. □

5 Upgrade attempts, novelty, and scope

The first route used an ordinary average over a positive-measure set. It fails because the sum of two pointwise distances does not control the sum of their two essential suprema. A second route worked when X was finite-dimensional: an ultrafilter limit of every bounded X -valued function then lies in the compact unit ball, so one can patch near that limit. The decisive upgrade removes compactness. It uses a countably valued field of slice functionals and lets the Boolean ultrafilter select a *finite distance bin*; this controls both suprema without requiring an X -valued ultralimit. It also shows that no σ -finiteness assumption is needed.

A bounded primary-source search covered the exact source statement, “generalized-lush”, “GL-space”, “ $L_\infty(\mu, X)$ ”, and later arXiv citations. It found the source, the original paper proving stability under $C(K, X)$ and ℓ_∞ -sums, and work on property (**), but no primary paper proving this implication. Novelty confidence is moderate; correctness confidence is high.

The proof resolves only the forward implication from X to $L_\infty(\mu, X)$. It does not show that GL descends from $C(K, X)$, $L_1(\mu, X)$, or $L_\infty(\mu, X)$ to a nonseparable X . Direct restriction and averaging attempts at those converses lose the single slice functional that must work simultaneously for the entire nonseparable sphere. Those questions remain outside the claim of this packet.

References

- [1] K. Zhao and D. Tan, *A property in vector-valued function spaces*, arXiv:2007.10589.